Using Gauge Pressure to Calculate the Height of a Water Column

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PURPOSE

The purpose of this investigation is to demonstrate the relationship between atmospheric pressure and the height of the water column in a closed system. This is achieved by creating a partial vacuum at the top of the column, which allows atmospheric pressure to push the water up the tube. The resulting gauge pressure of the column is measured with a manometer, allowing the height of the column to be calculated using the hydrostatic equation ($P = \rho gh$). The pressure difference between the atmospheric pressure and the pressure inside the column can be quantified, as it is proportional to the height of the water column. This experiment verifies the theoretical predictions of hydrostatic equilibria in a closed fluid system.

THEORY

In a static fluid, pressure will increase with depth. The hydrostatic pressure equation describes this relationship:

$$P = P_0 + \rho gh$$

where ρ is the fluid's density, g is the gravitational acceleration, h is the depth, and P_0 is the surface pressure. When a partial vacuum is created at the top of the tube, the internal pressure is reduced below atmospheric pressure. This pressure imbalance causes atmospheric pressure to act on the surface of the reservoir, sending water up into the column. Water rises until this equilibrium is reached, shown in this relationship:

$$P_{atm} - P_{column} = \rho gh$$

This pressure difference is referred to as the gauge pressure, which is defined as the difference between atmospheric pressure and the column's internal pressure:

$$P_{gauge} = \rho g h$$

For these experimental conditions, the density of water is $\rho = 1000\ kg/m^3$, Earth's gravitational pull is $g = 9.80\ m/s^2$, and h is the water column height in meters. The gauge pressure is measured with a manometer, enabling computation of the column height. This value can be compared with the measured value of the tube height, verifying the theoretical relationship between atmospheric pressure and fluid column height.

EXPERIMENTAL

Equipment Setup

A sealed column was partially submerged in a water reservoir. The column has a valve at the top with a pneumatic connection that allows air to be pulled out of the tube (Figure 1). This valve has a lever that enables the temporary connection of a manometer for measuring the current internal gauge pressure (Figure 2).

Figure 1: Pneumatic connection between column and pump.

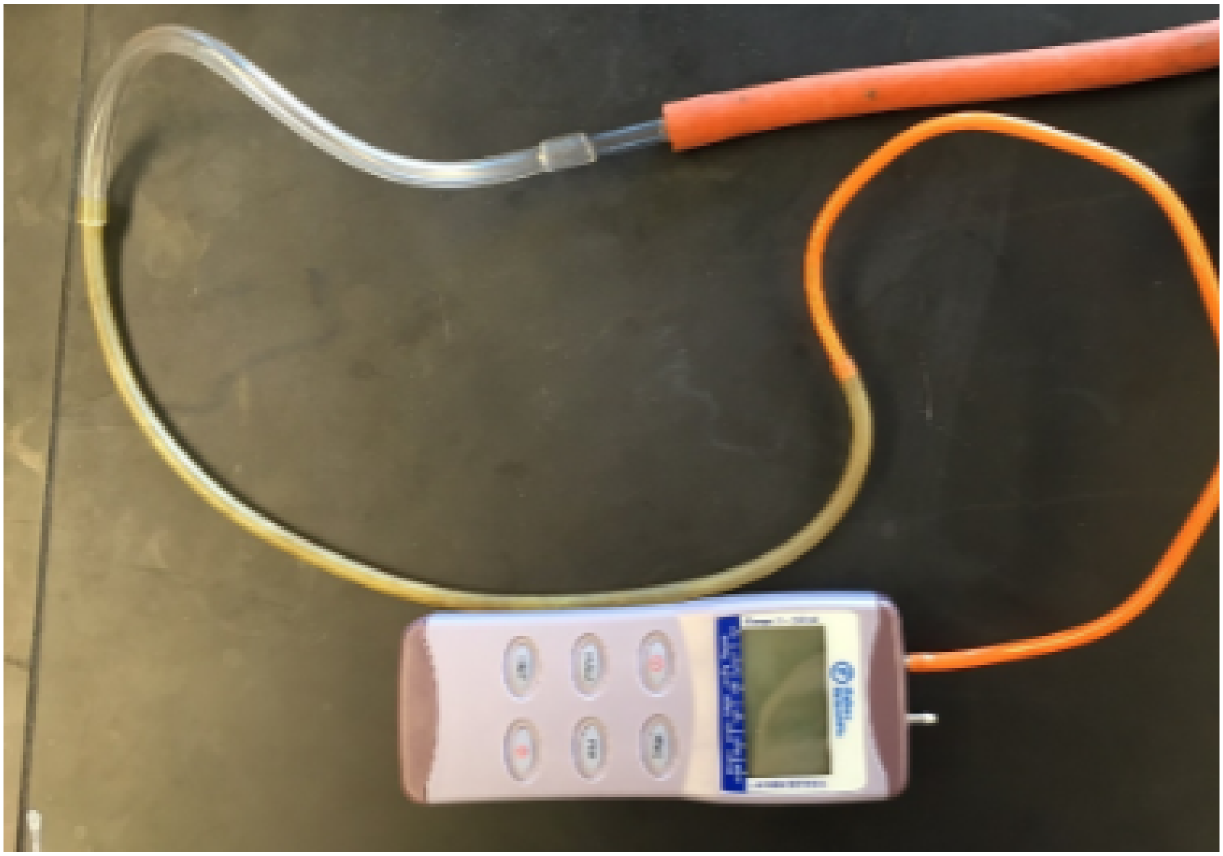

Figure 2: Connection for pressure sensor.

Experimental Procedure

The current atmospheric pressure was measured using a barometer, and the diameter of the column was measured using a meter stick. The vacuum pump was attached to the valve on top of the column and was activated until the water level reached ¾ of the column. The vacuum was switched off, and the manometer was attached to measure the gauge pressure. The water was allowed to drain to ½ and ¼, where both gauge pressures were recorded.

DATA

Reported Data

For this experiment, the theoretical heights given the gauge pressure were calculated. The experimental conditions are reported in Table 1, and the measured data and results are reported in Table 2.

Diameter (m)	0.13
Atmospheric Pressure (mm Hg)	760.48
Temperature (C°)	22

Table 1: Experimental Parameters

Percent of Water Column Filled	Calculated Height of Column (m)	Measured Height of Column (m)	Percent Error of Heights	Gauge Pressure of Column (mm Hg)
75%	1.05	1.01	3.6%	77
50%	0.76	0.73	3.8%	56
25%	0.38	0.36	4.7%	28

Table 2: Measured Values and Calculated Results of the Experiment

Example Calculation

Starting with the hydrostatic pressure equation and rearranging for height:

$$P_{gauge} = \rho g h$$

$$h = \frac{P_{gauge}}{\rho g}$$

Measurements taken in mm Hg, convert to torr/cm. Plug in constants:

$$h = \frac{P_{gauge}}{1000\frac{kg}{m^3} * 9.8\frac{m}{s^2}} * 100\frac{cm}{m} * \frac{1torr}{133.322pa}$$

Final conversion factor:

$$h = \frac{p}{0.73506\frac{torr}{cm}}$$

Plug in measured pressure (example for ¾ full column):

$$h = \frac{77mmhg}{0.73506\frac{torr}{cm}} = 104.753cm$$

Calculate percent error:

$$\frac{Theoretical - Experimental}{Theoretical} * 100 = Percent\ Error$$

$$\frac{104.753cm - 101cm}{104.753cm} = 3.58\%$$

Percent error on ¾ full column = 3.58%

Data Work-up

Calculated data values in wxMaxima:

```
(%i7)   rho:1000;  /* Define rho as a constant */
rho     1000
(%i8)   g:9.80; /* Define gravity as a constant */
g       9.8
(%i9)   patotorr:1/133.322; /* Conversion factor for Pascals to Torr */
patotorr 0.007500637554192106
(%i10)  mtocm:100; /* Conversion factor for m to cm */
mtocm   100
(%i11)  rhog:rho·g·patotorr/mtocm; /* Calculating rho*g in torr/cm */
rhog    0.7350624803108264
(%i12)  h(p):=p/rhog /* Define H as a function of pressure/conversion factor */;
(%o12)  h(p):= p/rhog
(%i16)  hColumnP:[77,56,28]; /* Measured Pressures */
hColumnP [77,56,28]
(%i14)  nhColumnP:length(hColumnP); /* Number of Trials */
nhColumnP 3
(%i17)  map(h, hColumnP); /*Maps measured pressures to function H */
(%o17)  [104.753,76.184,38.092]
(%i21)  hColumnMeasured:[101,73.3,36.3]; /* Measured experimental column height in cm */
hColumnMeasured [101,73.3,36.3]
(%i20)  (%o17 − hColumnMeasured)/%o17 · 100; /*Percent error of the values */
(%o20)  [3.582713621566924,3.7855717736007564,4.704399873989294]
```

RESULTS

From the experiment it was found that the measured pressure calculated through the hydrostatic equation produced height values similar to those measured in the trial. Shown in Table 2, the percent error on average has a value of 4.03%. Some possible sources of error include the leaking of the column when changing the pneumatic connection from the pump to the manometer. Another possible source of error is the column shape, the column thickness could have impacted the height measurement taken by a few centimeters. Overall the calculated values are consistent with the measured height values of the column.

REFERENCES

East Carolina University, Department of Chemistry. (2022). Physical Chemistry I Laboratory CHEM 3951 [Laboratory manual].

SUPPLEMENTAL INFORMATION

See attached Maxima PDF file: **Physical Chemistry Lab 1 Workup**